

Temporal Early-Warning Experiment Results

OULAD withdrawal prediction using cumulative day 35, day 60, and day 75 observation windows

Controlled comparison. The same 32,593 records, target, 60/20/20 split membership, preprocessing steps, XGBoost hyperparameters, and held-out test set were used at every window. Only the assessment and VLE activity available by each cutoff changed.

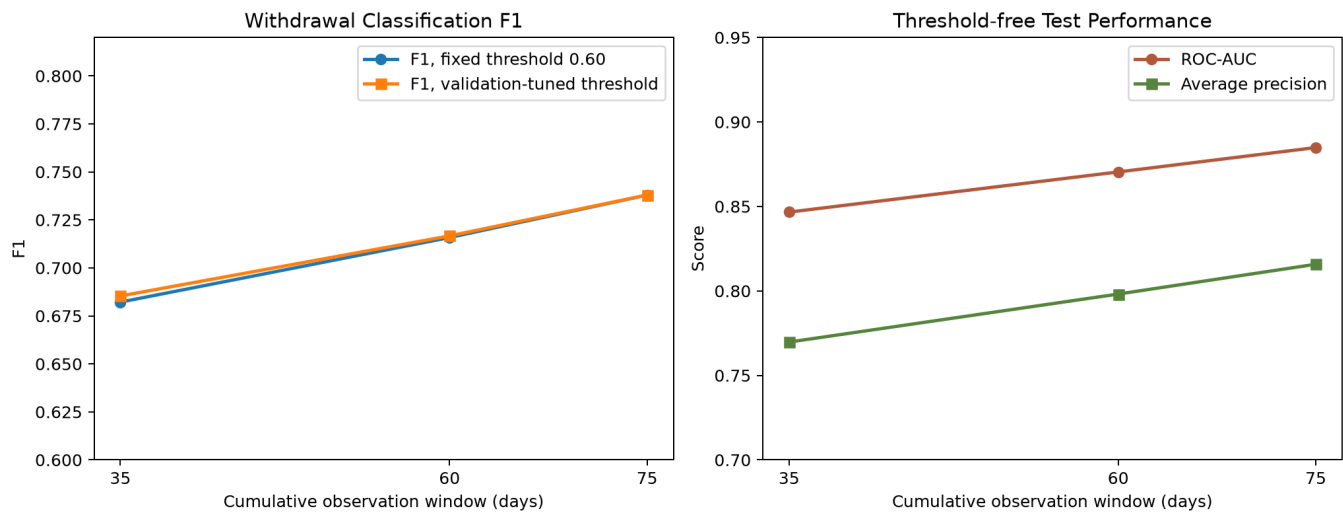
Window	Validation threshold	Test F1 at 0.60 [95% CI]	ROC-AUC [95% CI]	Average precision	5-fold CV F1 mean +/- SD
Day 35	0.520	0.682 [0.665, 0.699]	0.847 [0.837, 0.857]	0.770	0.689 +/- 0.011
Day 60	0.560	0.716 [0.699, 0.731]	0.870 [0.860, 0.880]	0.798	0.718 +/- 0.010
Day 75	0.600	0.738 [0.723, 0.753]	0.885 [0.876, 0.894]	0.816	0.736 +/- 0.008

Main result: fixed-threshold test F1 increased from 0.682 at day 35 to 0.716 at day 60 and 0.738 at day 75. ROC-AUC increased from 0.847 to 0.885. The day-75 values exactly reproduce the submitted Milestone 5 OULAD result.

Primary comparison threshold: 0.60, selected in Milestone 5. Confidence intervals use 1,000 bootstrap samples of the shared held-out test set. SHAP uses the same reproducible sample of 500 test records at each window.

Performance Across Time

Both threshold-based and threshold-free measures improve as additional course activity becomes available.



Paired bootstrap differences use the same resampled test students for both windows. Positive values favor the later window.

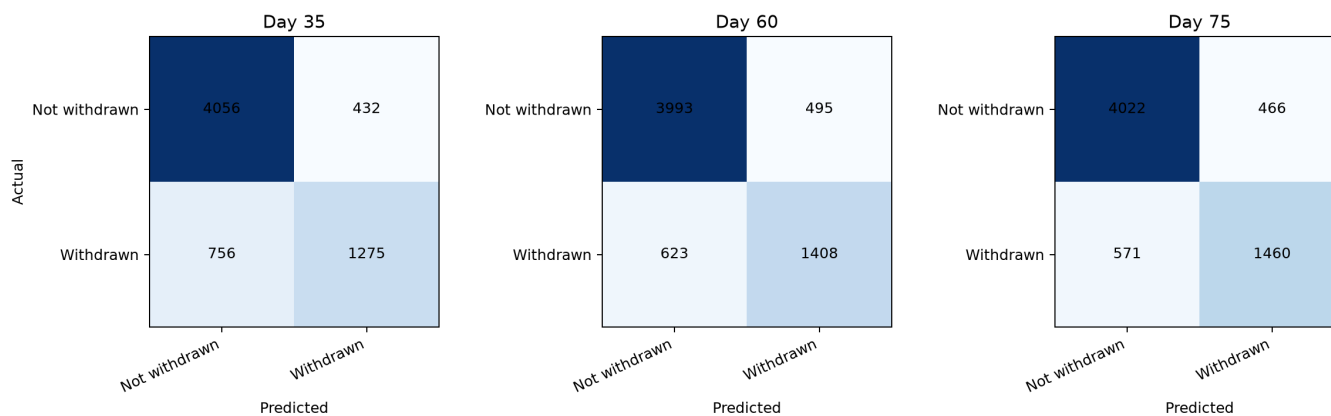
Comparison	Metric	Mean change	95% CI	Excludes zero
Day 60 - Day 35	F1	+0.034	[+0.023, +0.045]	Yes
Day 60 - Day 35	ROC_AUC	+0.024	[+0.018, +0.029]	Yes
Day 75 - Day 60	F1	+0.022	[+0.014, +0.031]	Yes
Day 75 - Day 60	ROC_AUC	+0.015	[+0.010, +0.019]	Yes
Day 75 - Day 35	F1	+0.056	[+0.044, +0.068]	Yes
Day 75 - Day 35	ROC_AUC	+0.038	[+0.032, +0.045]	Yes

All paired 95% intervals are above zero. Within this fixed test split, the later windows produce measurable improvements rather than only small point-estimate changes.

Classification Results

The fixed 0.60 threshold is used below so the decision rule does not change between windows.

Held-out Test Confusion Matrices at Fixed Threshold 0.60

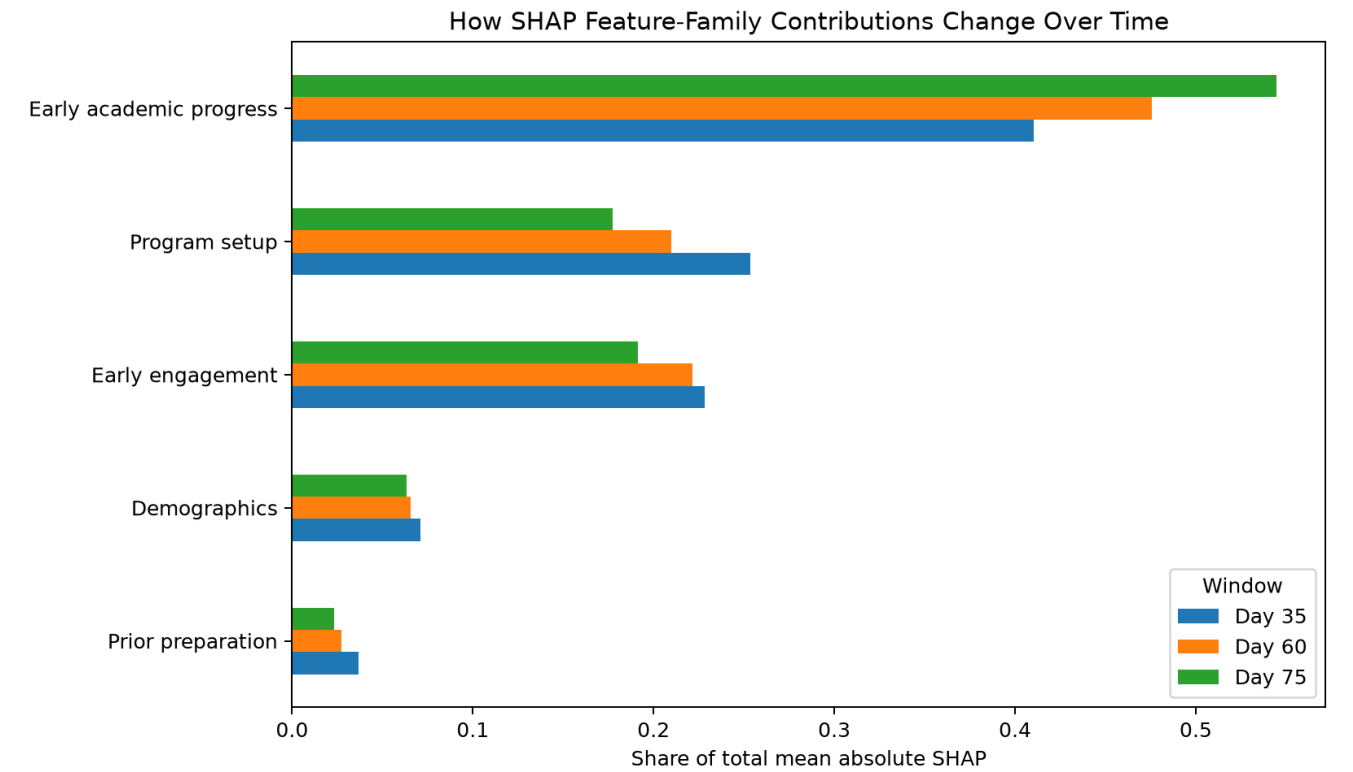


Window	Accuracy	Balanced accuracy	Precision	Recall	F1	Validation-selected threshold	F1 at selected threshold
Day 35	0.818	0.766	0.747	0.628	0.682	0.520	0.685
Day 60	0.829	0.791	0.740	0.693	0.716	0.560	0.717
Day 75	0.841	0.808	0.758	0.719	0.738	0.600	0.738

Operational reading: the fixed-threshold withdrawal recall rises from 0.628 at day 35 to 0.719 at day 75 while precision remains near 0.75. The model therefore identifies more withdrawn students as their assessment history develops without a large loss of alert precision.

SHAP Feature-Family Shift

Shares are calculated from total mean absolute SHAP. They show how the model's explanation mix changes, not causal effects.

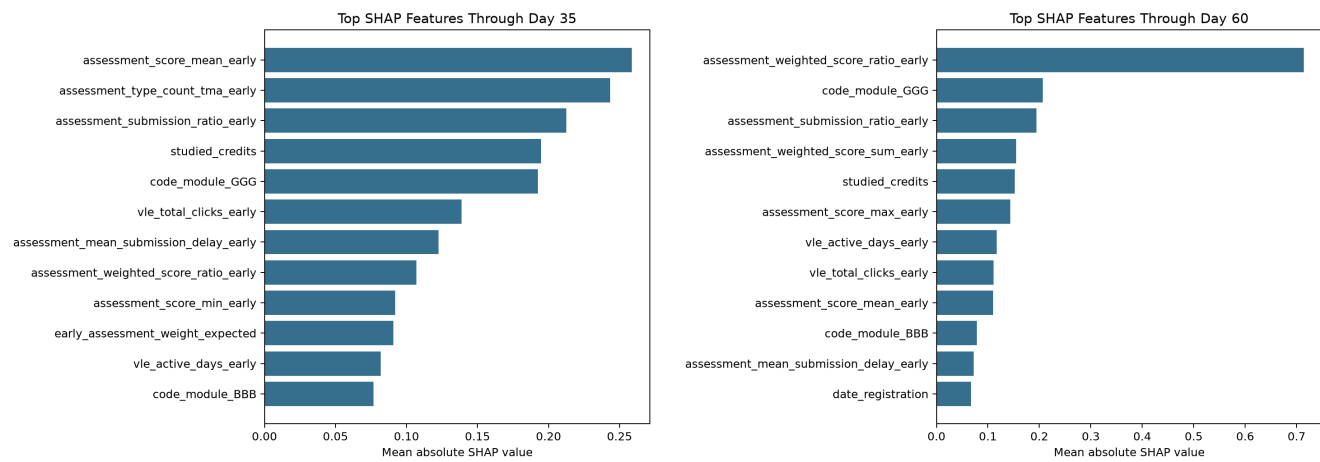


Feature family	Day 35	Day 60	Day 75
Early academic progress	41.0%	47.6%	54.5%
Early engagement	22.8%	22.1%	19.1%
Program setup	25.3%	21.0%	17.8%
Demographics	7.1%	6.6%	6.3%
Prior preparation	3.7%	2.7%	2.3%

Academic progress becomes more influential over time (41.0% to 54.5%). Program setup and prior preparation decline as the model receives direct evidence of current-course performance. Engagement remains useful at every window.

Window-Specific SHAP Results

Day 35 and day 60 use the same fitted model design but explain different amounts of observed activity.

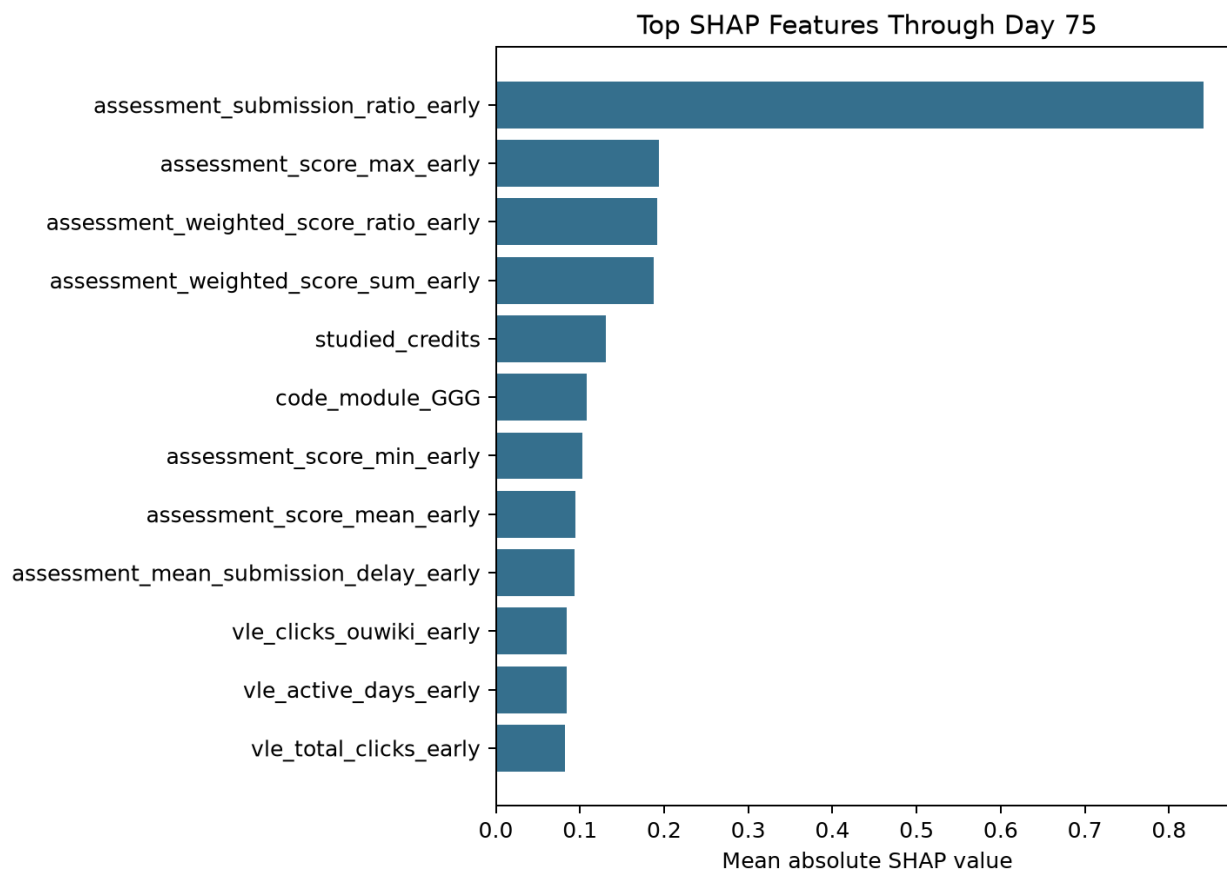


Window	Leading SHAP signals	Result-oriented reading
Day 35	Assessment score mean, TMA count, submission ratio	The earliest alerts depend on first assessment participation and initial scores.
Day 60	Weighted score ratio, submission ratio, weighted score sum	By day 60, accumulated assessment performance provides a clearer risk signal.

Intervention link: the day-35 model supports early reminders and first-assessment feedback. By day 60, outreach can focus more directly on low weighted performance and missing work.

Day 75 SHAP Results

The day-75 experiment reproduces the submitted Milestone 5 OULAD model and completes the temporal comparison.



Rank	Feature	Mean absolute SHAP
1	assessment_submission_ratio_early	0.841
2	assessment_score_max_early	0.194
3	assessment_weighted_score_ratio_early	0.192
4	assessment_weighted_score_sum_early	0.188
5	studied_credits	0.131

By day 75, assessment submission ratio is the strongest SHAP signal. This supports targeted follow-up for students who are not completing the assessments available by that stage. SHAP describes model contribution and should not be presented as proof that a feature causes withdrawal.